Online Appendix to
"Target-Population Fragility in Judge IV under Average Monotonicity"

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This appendix supports the main text. Appendix A proves the three-judge results of Section 3 and records the full response-type table. Appendix B extends the boundary mechanism to any finite number of judges. Appendix C derives the boundary diagnostic and constructs the Philadelphia propensities and standard errors. Appendix D reports simulations that validate the diagnostic as a finite-sample classification tool. Appendix E positions the paper in the related literature. Equation, proposition, and table numbers carry the appendix letter.

A Proofs for the Three-Judge Case

This appendix proves the two results stated in Section 3 and records the response-type table to which the main text refers. Judges are indexed by $z \in \{1, 2, 3\}$, ordered so that $p_1 \leq p_2 \leq p_3$, with assignment probabilities $\lambda_z > 0$ and assignment-weighted average propensity $\bar{p} = \sum_{z=1}^{3} \lambda_z p_z$. A response type is $g = (g_1, g_2, g_3) \in \{0,1\}^3$, and its type-specific weight is $\omega(g) = \sum_{z=1}^{3} \lambda_z (p_z - \bar{p}) g_z$.

A.1 Type-specific weights

The weight admits a representation relative to the intermediate judge. Because $\sum_{z=1}^{3} \lambda_z (p_z - \bar{p}) = 0$,

$$\omega(g) = \lambda_1(p_1 - \bar{p})(g_1 - g_2) + \lambda_3(p_3 - \bar{p})(g_3 - g_2), \tag{A.1}$$

where $\lambda_1(p_1 - \bar{p})$ is nonpositive and $\lambda_3(p_3 - \bar{p})$ is nonnegative because $p_1 \leq \bar{p} \leq p_3$. The eight weights follow from (A.1). The always-taker and never-taker types receive zero weight, $\omega(1,1,1) = \omega(0,0,0) = 0$. The monotone compliers receive nonnegative weight,

$$\omega(0,0,1) = \lambda_3(p_3 - \bar{p}) \geq 0, \qquad \omega(0,1,1) = -\lambda_1(p_1 - \bar{p}) \geq 0.$$

The reverse types receive nonpositive weight,

$$\omega(1,1,0) = -\lambda_3(p_3 - \bar{p}) \leq 0, \qquad \omega(1,0,0) = \lambda_1(p_1 - \bar{p}) \leq 0.$$

The two non-monotone mirror types satisfy

$$\omega(1,0,1) = \lambda_1(p_1 - \bar{p}) + \lambda_3(p_3 - \bar{p}) = -\lambda_2(p_2 - \bar{p}), \qquad \omega(0,1,0) = \lambda_2(p_2 - \bar{p}).$$

These weights populate Table A.1.

Table A.1: Response Types, IV Weights, and Admissibility

Type	(D_{i1}, D_{i2}, D_{i3})	$\omega(g)$	Strong monotonicity	Average monotonicity
g_1	$(1,1,1)$	0	Yes	Yes
g_2	$(0,0,0)$	0	Yes	Yes
g_3	$(0,0,1)$	$\lambda_3(p_3 - \bar{p}) > 0$	Yes	Yes
g_5	$(0,1,1)$	$-\lambda_1(p_1 - \bar{p}) > 0$	Yes	Yes
g_6	$(1,1,0)$	$-\lambda_3(p_3 - \bar{p}) < 0$	No	No
g_7	$(1,0,0)$	$\lambda_1(p_1 - \bar{p}) < 0$	No	No
g_4	$(1,0,1)$	$-\lambda_2(p_2 - \bar{p})$	No	Depends
g_8	$(0,1,0)$	$\lambda_2(p_2 - \bar{p})$	No	Depends

Notes: The table reports response types in the three-judge case, ordered by judge propensities $p_1 \leq p_2 \leq p_3$. The signs for g_3, g_5, g_6, and g_7 are strict when $p_1 < \bar{p} < p_3$. At boundary cases, the corresponding weights may be zero. Under average monotonicity, a response type is admissible only if its weight is nonnegative.

A.2 Proof of Proposition 1

Proof of Proposition 1. The two mirror types carry the only weights whose sign depends on $p_2 - \bar{p}$. Their weights are $\omega(g_4) = -\lambda_2(p_2 - \bar{p})$ and $\omega(g_8) = \lambda_2(p_2 - \bar{p})$, and $\lambda_2 > 0$. When $p_2 < \bar{p}$, $\omega(g_4) > 0$ and $\omega(g_8) < 0$, so average monotonicity admits g_4 and rules out g_8. When $p_2 > \bar{p}$, the signs reverse, so average monotonicity admits g_8 and rules out g_4. When $p_2 = \bar{p}$, both weights are zero. All other response types have signs that do not depend on $p_2 - \bar{p}$, so the only change in the positive-weight set as p_2 crosses $\bar{p}$ is the switch between g_4 and g_8. The positive-weight set therefore changes discontinuously at $p_2 = \bar{p}$. ■

A.3 Proof of Corollary 1

Proof of Corollary 1. Write the IV estimand under average monotonicity as a ratio over positively weighted types,

$$\beta^{IV} = \frac{\sum_{g:\omega(g)>0} \pi_g \, \omega(g) \, \tau_g}{\sum_{g:\omega(g)>0} \pi_g \, \omega(g)},$$

where $\pi_g = \Pr(G_i = g)$ and $\tau_g = E[\delta_i \mid G_i = g]$. Zero-weight types do not enter.

The scalar estimand is continuous. First fix the population $\{\pi_g\}_g$ and vary p_2. Every weight $\omega(g)$ is continuous in p_2, and the switching weights $\omega(g_4) = -\lambda_2(p_2 - \bar{p})$ and $\omega(g_8) = \lambda_2(p_2 - \bar{p})$ vanish as $p_2 \to \bar{p}$. Both numerator and denominator of β^{IV} are therefore continuous in p_2, and wherever the denominator is bounded away from zero β^{IV} is continuous. For a fixed population there is thus no jump in the scalar estimand at the boundary.

The admissible support is discontinuous. By Proposition 1 the average-monotonicity-admissible set switches g_4 for g_8 as p_2 crosses $\bar{p}$: on the side $p_2 < \bar{p}$ a positive share on g_8 would give it the negative weight $\omega(g_8) < 0$, and on the side $p_2 > \bar{p}$ a positive share on g_4 would give $\omega(g_4) < 0$. Preserving average monotonicity across the boundary therefore requires two distinct populations, one containing g_4 and one containing its mirror g_8. Their identified parameters,

$$\beta_- = \frac{\sum_{g \in \{g_3, g_4, g_5\}} \pi_g \, \omega(g) \, \tau_g}{\sum_{g \in \{g_3, g_4, g_5\}} \pi_g \, \omega(g)}, \qquad \beta_+ = \frac{\sum_{g \in \{g_3, g_5, g_8\}} \pi_g \, \omega(g) \, \tau_g}{\sum_{g \in \{g_3, g_5, g_8\}} \pi_g \, \omega(g)},$$

average over disjoint switching components and need not coincide. The example in Appendix A.4 gives admissible shares and heterogeneous effects with $\tau_{g_4} \neq \tau_{g_8}$ for which $\beta_- \neq \beta_+$. Hence two arbitrarily close average-monotonicity-consistent designs can identify effects for different groups, even though within either design β^{IV} moves continuously. ■

A.4 A worked example of local non-invariance

The lack of guaranteed local invariance is not merely formal. The following example shows that two average-monotonicity-consistent designs arbitrarily close in the propensity profile can identify effects for disjoint groups.

Example A.1 (Target population across the boundary)**.** *Consider two three-judge designs that differ only in whether p_2 lies just below or just above $\bar{p}$, each consistent with average monotonicity. In the first, $p_2 < \bar{p}$ and the non-monotone type present is $g_4 = (1, 0, 1)$; in the second, $p_2 > \bar{p}$ and it is the mirror $g_8 = (0, 1, 0)$. Apart from the switching type, both populations contain only always-takers and never-takers, which carry zero weight. Average monotonicity holds in each design, since the admitted non-monotone type carries positive weight while its mirror is absent. Set $\tau_{g_4} \neq \tau_{g_8}$. The only positively weighted type is the switching type, so the first design identifies $\beta_- = \tau_{g_4}$ and the second $\beta_+ = \tau_{g_8}$. The two designs are arbitrarily close in the propensity profile yet identify effects for disjoint groups, so $\beta_- \neq \beta_+$. The discontinuity is in which group the estimand represents; for a single fixed population, β^{IV} would vary continuously in p_2.*

B General Finite-J Extension

This appendix shows that the boundary mechanism of Section 3 is not specific to three judges. Judges are indexed by $z \in \{1, \ldots, J\}$ with $J \geq 3$, each assigned with probability $\lambda_z > 0$, and each individual has a response type $\tau = (\tau_1, \ldots, \tau_J) \in \{0,1\}^J$. The assignment-weighted average propensity is $\bar{p} = \sum_{z=1}^{J} \lambda_z p_z$.

The type-specific weight is an inner product between the response type and the propensity-deviation vector. Let $v = (\lambda_1(p_1 - \bar{p}), \ldots, \lambda_J(p_J - \bar{p}))$. Then

$$\omega(\tau) = \sum_{z=1}^{J} \lambda_z (p_z - \bar{p}) \tau_z = \langle v, \tau \rangle, \qquad \langle v, \mathbf{1} \rangle = 0,$$

where $\mathbf{1} = (1, \ldots, 1)$, since $\sum_{z=1}^{J} \lambda_z (p_z - \bar{p}) = 0$. Every response type has a componentwise complement with the opposite weight: for any $\tau \in \{0,1\}^J$,

$$\omega(\mathbf{1} - \tau) = \langle v, \mathbf{1} - \tau \rangle = \langle v, \mathbf{1} \rangle - \langle v, \tau \rangle = -\omega(\tau).$$

This complementarity is the finite-J version of the g_4-g_8 relationship in the three-judge case.

A single judge crossing the boundary changes the positive-weight set. Fix judge j and let $e_j = (0, \ldots, 0, 1, 0, \ldots, 0)$ be the type treated only by judge j. Its complement $\mathbf{1} - e_j$ is the type treated by every judge except j. Their weights are

$$\omega(e_j) = \lambda_j (p_j - \bar{p}), \qquad \omega(\mathbf{1} - e_j) = -\lambda_j (p_j - \bar{p}),$$

which switch sign exactly when p_j crosses $\bar{p}$.

Proposition B.1 (General boundary discontinuity)**.** *Fix a judge j and let $\mathcal{A}(v) = \{\tau \in \{0,1\}^J : \langle v, \tau \rangle \geq 0\}$ denote the set of response types with nonnegative average-monotonicity weight. At any propensity profile satisfying $p_j = \bar{p}$, the set-valued map $v \mapsto \mathcal{A}(v)$ changes discontinuously: for two arbitrarily close profiles, one with $p_j < \bar{p}$ and one with $p_j > \bar{p}$, the admissible sets differ by at least the two types e_j and $\mathbf{1} - e_j$.*

Proof. If $p_j < \bar{p}$, then $\omega(e_j) < 0$ and $\omega(\mathbf{1} - e_j) > 0$, so $e_j \notin \mathcal{A}(v)$ while $\mathbf{1} - e_j \in \mathcal{A}(v)$. If $p_j > \bar{p}$, then $\omega(e_j) > 0$ and $\omega(\mathbf{1} - e_j) < 0$, so $e_j \in \mathcal{A}(v)$ while $\mathbf{1} - e_j \notin \mathcal{A}(v)$. At $p_j = \bar{p}$, both types have zero weight. The admissible set therefore changes discretely when judge j crosses the boundary, and since such profiles can be chosen arbitrarily close to the boundary, $v \mapsto \mathcal{A}(v)$ is discontinuous at $p_j = \bar{p}$. $\square$

The finite-J estimand inherits the same target-population problem. With $\pi_\tau = \Pr(G_i =$

τ) and $\gamma_\tau = E[\delta_i \mid G_i = \tau]$,

$$\beta^{IV}(v) = \frac{\sum_{\tau:\langle v,\tau\rangle>0} \pi_\tau \langle v, \tau\rangle \gamma_\tau}{\sum_{\tau:\langle v,\tau\rangle>0} \pi_\tau \langle v, \tau\rangle}.$$

If e_j and $\mathbf{1} - e_j$ have positive population shares and different average effects, crossing the boundary changes which treatment-effect component enters the estimand. The scope for such switches grows with the number of judges: any judge whose propensity lies near $\bar{p}$ can generate a pairwise switch, and several judges near the boundary can be weakly classified at once. This is why the diagnostic of Section 4 reports judge-level distances, assignment shares, and a design-level minimum distance.

C The Boundary Diagnostic: Propensities, Standard Errors, and Sign Classification

This appendix derives the boundary diagnostic of Section 4 and constructs the objects used in the Philadelphia application. The diagnostic is a sign-classification statistic for each judge's position relative to the average-monotonicity boundary, not a test of average monotonicity.

C.1 Population object and Wald interpretation

The population object is the judge-specific distance from the boundary, $d_z = p_z - \bar{p}$ with $\bar{p} = \sum_{j=1}^{J} \lambda_j p_j$. Its sign determines whether judge z lies above or below the boundary, and under average monotonicity that sign governs which mirror response types can receive positive weight. The sample analogue is $\hat{d}_z = \hat{p}_z - \hat{\bar{p}}$ with $\hat{\bar{p}} = \sum_{j=1}^{J} \hat{\lambda}_j \hat{p}_j$, and the standardized diagnostic is $\hat{r}_z = |\hat{d}_z| / \widehat{\mathrm{se}}(\hat{d}_z)$. Direction is given by $\mathrm{sign}(\hat{d}_z)$ and distance from the boundary by $\hat{r}_z$.

Lemma C.1 (Wald interpretation)**.** *For any cutoff $c > 0$,*

$$\hat{r}_z \le c \quad \Longleftrightarrow \quad 0 \in \left[\hat{d}_z - c\,\widehat{\mathrm{se}}(\hat{d}_z),\ \hat{d}_z + c\,\widehat{\mathrm{se}}(\hat{d}_z)\right].$$

Proof. By definition $\hat{r}_z \le c$ is equivalent to $|\hat{d}_z| \le c\,\widehat{\mathrm{se}}(\hat{d}_z)$, which holds if and only if $\hat{d}_z - c\,\widehat{\mathrm{se}}(\hat{d}_z) \le 0 \le \hat{d}_z + c\,\widehat{\mathrm{se}}(\hat{d}_z)$, that is, zero lies in the displayed Wald interval. □

The Wald interpretation links the diagnostic to target-population ambiguity. When $\hat{r}_z \le c$, the sample does not separate judge z from the boundary at cutoff c, and because average-monotonicity weights depend on the sign of d_z, a weakly pinned sign means a weakly pinned positive-weight classification.

C.2 Local-to-boundary sign error

A local-to-boundary approximation gives the diagnostic its sign-error reading. Consider a sequence of designs indexed by n, with population distance $d_{z,n}$, estimator $\hat{d}_{z,n}$, and standard error $\sigma_{z,n} = \mathrm{se}(\hat{d}_{z,n})$. Suppose

$$\frac{\hat{d}_{z,n} - d_{z,n}}{\sigma_{z,n}} \xrightarrow{d} \mathcal{N}(0,1), \qquad \frac{d_{z,n}}{\sigma_{z,n}} \to \rho_z,$$

with ρ_z finite and nonzero. Then

$$\Pr\Big(\mathrm{sign}(\hat{d}_{z,n}) \neq \mathrm{sign}(d_{z,n})\Big) \to \Phi(-|\rho_z|).$$

The probability of sign misclassification is therefore governed by distance to the boundary in standard-error units, which is what $\hat{r}_z$ estimates. The design-level summary is the minimum standardized distance $\hat{r}_{\min} = \min_z \hat{r}_z$: a high value means every judge is separated from the boundary, while a low value means at least one judge is weakly classified. The minimum is an alarm statistic only and does not replace the full diagnostic vector $\{\hat{d}_z, \hat{r}_z, \hat{\lambda}_z\}_{z=1}^{J}$, since the identity of the judge attaining $\hat{r}_{\min}$ can itself be unstable when several judges lie near the boundary.

C.3 Philadelphia propensities and standard errors

The Philadelphia diagnostic uses the maximum-control specification of the Stevenson replication. The first-stage outcome is pretrial detention, the excluded instruments are the eight judge indicators, and the controls include charge categories, demographics, prior-record variables, offense indicators, and court-calendar controls. The judge propensities $\hat{p}_z$ are regression-adjusted detention margins, computed by setting the judge indicator to judge z and holding the control distribution fixed at the sample distribution, and $\hat{\lambda}_z$ is the empirical assignment share. The boundary distance is $\hat{d}_z = \hat{p}_z - \hat{\bar{p}}$ with $\hat{\bar{p}} = \sum_{z=1}^{J} \hat{\lambda}_z \hat{p}_z$.

The standard errors treat empirical assignment shares as fixed. Let $\widehat{V}_p$ denote the estimated covariance matrix of the adjusted judge margins $\hat{p} = (\hat{p}_1, \ldots, \hat{p}_J)'$, and define $A = I_J - \mathbf{1}\hat{\lambda}'$ with $\hat{\lambda} = (\hat{\lambda}_1, \ldots, \hat{\lambda}_J)'$. Then the vector of boundary distances is $\hat{d} = A\hat{p}$, its delta-method covariance is $\widehat{V}_d = A\widehat{V}_p A'$, and the standard error for judge z is $\widehat{\mathrm{se}}(\hat{d}_z) = \sqrt{e_z' \widehat{V}_d e_z}$, where e_z is the z-th unit vector. A judge is boundary-fragile at cutoff c when $\hat{r}_z \leq c$, and

Section 4 uses $c = 1.96$.

Table C.1 reports the full judge-level diagnostics. The average propensity in the maximum-control specification is $\hat{\bar{p}} = 0.4116$. At $c = 1.96$, Judges 1, 5, and 7 are boundary-fragile, with combined assignment share 0.291. Judge 7 is closest to the boundary, with $\hat{d}_7 = 0.0009$ and $\hat{r}_7 = 0.44$.

Table C.1: Judge-level boundary diagnostics in the Philadelphia bail data

Judge	Fragile	Cases	$\hat{\lambda}_z$	$\hat{p}_z$	$\hat{d}_z$	$\widehat{\mathrm{se}}(\hat{d}_z)$	$\hat{r}_z$	Side
1	Yes	21,523	0.065	0.4071	-0.0044	0.0031	1.45	Below
2	No	13,087	0.039	0.4378	0.0262	0.0041	6.37	Above
3	No	54,272	0.163	0.4186	0.0070	0.0017	4.21	Above
4	No	56,585	0.170	0.3936	-0.0179	0.0016	11.15	Below
5	Yes	33,690	0.101	0.4077	-0.0039	0.0023	1.69	Below
6	No	55,038	0.166	0.4304	0.0188	0.0017	11.28	Above
7	Yes	41,475	0.125	0.4124	0.0009	0.0020	0.44	Above
8	No	56,301	0.170	0.4017	-0.0098	0.0017	5.94	Below
Average propensity $\hat{\bar{p}}$					0.4116			
Minimum raw distance $\min_z \lvert\hat{d}_z\rvert$					0.0009			
Minimum standardized distance $\hat{r}_{\min}$					0.44			
Boundary-fragile judges under $c = 1.96$					1, 5, 7			
Total assignment share of fragile judges					0.291			
Total cases					331,971			

Notes: A judge is classified as boundary-fragile when $\hat{r}_z \leq 1.96$. The propensity $\hat{p}_z$ is the regression-adjusted judge-specific detention propensity from the first-stage equation. Standard errors for $\hat{d}_z$ are computed by the delta method using the covariance matrix of the adjusted judge margins, treating empirical assignment shares as fixed.

D Simulations

This appendix reports simulations that validate the boundary diagnostic as a finite-sample classification tool. The diagnostic asks whether the data place each judge securely on one side of the average-monotonicity boundary. The simulations check three things implied by the theory: that low $\hat{r}_z$ corresponds to high rates of sign misclassification, that fragility becomes more common as the number of judges grows, and that the diagnostic-based boundary interval does not exclude the true boundary in finite samples.

D.1 Design

The data-generating process starts from a finite judge design. Judge z has treatment propensity $p_z = E[D \mid Z = z]$ and assignment share $\lambda_z = P(Z = z)$, with $\sum_z \lambda_z = 1$, baseline boundary $\bar{p}^0 = \sum_z \lambda_z p_z$, and population distance $d_z^0 = p_z - \bar{p}^0$. Each individual belongs to a response type g with treatment profile $D(g) = \{D_1(g), \ldots, D_J(g)\}$ and type share π_g, so that $p_z = \sum_g \pi_g D_z(g)$. In each replication individuals are drawn from the type distribution, judges are assigned according to λ, and treatment is $D_i = D_{Z_i}(g_i)$. Potential outcomes are drawn at the response-type level as independent Bernoulli variables, $Y_i(0) \sim \text{Bernoulli}(q_{0g})$ and $Y_i(1) \sim \text{Bernoulli}(q_{1g})$ with $\tau_g = q_{1g} - q_{0g}$, and the observed outcome is $Y_i = Y_i(0) + D_i\{Y_i(1) - Y_i(0)\}$. The baseline population estimand β^0 is fixed within each design; sampling variation enters only through the estimated objects $\hat{p}_z$, $\hat{d}_z$, $\hat{r}_z$.

D.2 Sample objects

Each replication estimates judge-level treatment and outcome means, $\hat{p}_z = n_z^{-1} \sum_{i:Z_i=z} D_i$, $\hat{y}_z = n_z^{-1} \sum_{i:Z_i=z} Y_i$, and $\hat{\lambda}_z = n_z/n$. The estimated boundary and distance are $\hat{\bar{p}} = \sum_z \hat{\lambda}_z \hat{p}_z$ and $\hat{d}_z = \hat{p}_z - \hat{\bar{p}}$, and the standardized diagnostic is $\hat{r}_z = |\hat{d}_z|/\widehat{\text{se}}(\hat{d}_z)$. In the covariate-free simulation the standard error uses the influence-function representation of $\hat{d}_z$,

$$\hat{\psi}_{iz} = \frac{\mathbf{1}\{Z_i = z\}}{\hat{\lambda}_z}(D_i - \hat{p}_z) - (D_i - \hat{\bar{p}}), \qquad \widehat{\text{se}}(\hat{d}_z) = \left[\tfrac{1}{n}\widehat{\text{Var}}(\hat{\psi}_{iz})\right]^{1/2}.$$

To calibrate the difficulty of the classification problem, we report the design signal-to-noise ratio $\rho_z = |d_z^0|/\text{sd}(\hat{d}_z)$, which the local theory of Appendix C.2 maps into a sign-error rate of approximately $\Phi(-\rho_z)$.

D.3 Designs

We study five designs, summarized in Table D.1. The two stylized three-judge designs isolate the boundary mechanism by placing one judge close to the boundary, one satisfying strong monotonicity and one satisfying average monotonicity without strong monotonicity. The Philadelphia-like design uses the empirical assignment shares and propensities from the application. The two bail designs are calibrated many-judge settings in the spirit of common judge and examiner applications. The many-judge designs are useful because the probability that at least one judge lies close to the boundary rises with the number of judges, even when no single judge is placed there by construction.

D.4 Results

The standardized boundary distance is informative about sign classification. Table D.2 groups judge-replication observations by $\hat{r}_z$ and reports the frequency with which the sample sign of $\hat{d}_z$ differs from the population sign of d_z^0. Sign errors concentrate among judges with

Table D.1: Population features of the simulation designs

Design	J	n	$\bar{p}^0$	$\min_z \lvert d_z^0 \rvert$	$\min_z \rho_z$
Stylized SM	3	10,000	0.498	0.0020	0.204
Stylized AM	3	10,000	0.498	0.0020	0.204
Philadelphia-like	8	20,000	0.4116	0.0008	0.091
Bail Philadelphia-like	9	50,000	0.560	0.0049	0.553
Bail Miami-like	60	50,000	0.560	0.0001	0.010

Notes: The stylized designs isolate the boundary mechanism in three-judge settings. The Philadelphia-like design uses the empirical assignment shares and treatment propensities from the application. The bail designs are calibrated many-judge designs. The statistic $\rho_z = |d_z^0|/\mathrm{sd}(\hat{d}_z)$ is the population signal-to-noise ratio for boundary classification.

low standardized distance and essentially disappear for large $\hat{r}_z$. In the Miami-like design, judges with $\hat{r}_z < 0.5$ are misclassified in 43 percent of samples, while judges with $\hat{r}_z > 4$ are never misclassified, and the same pattern holds across the Philadelphia-like designs.

Table D.2: Sign misclassification by standardized boundary distance

Design	$[0, 0.5)$	$[0.5, 1)$	$[1, 1.645)$	$[1.645, 1.96)$	$[1.96, 2.58)$	$[2.58, 4)$	$[4, \infty)$
Stylized SM	0.513	0.334	0.146	0.053	0.030	0.000	0.000
Stylized AM	0.392	0.284	0.096	0.050	0.020	0.000	0.000
Philadelphia-like	0.414	0.250	0.104	0.043	0.020	0.003	0.000
Bail Philadelphia-like	0.422	0.246	0.101	0.040	0.018	0.003	0.000
Bail Miami-like	0.426	0.243	0.100	0.039	0.019	0.003	0.000

Notes: Each entry reports the frequency with which the sample sign of $\hat{d}_z$ differs from the population sign of d_z^0, among judge-replication observations in the corresponding $\hat{r}_z$ bin. Sign errors are common when $\hat{r}_z$ is small and essentially disappear when $\hat{r}_z$ is large.

Boundary fragility is a design-level phenomenon in many-judge settings. Table D.3 reports the minimum standardized distance, the average number of fragile judges, the assignment share they handle, the probability of at least one sign error, and the frequency with which the plausible boundary interval contains the true boundary. The many-judge designs are markedly more fragile. In the Miami-like design with 60 judges, the average minimum standardized distance is close to zero, about 37 judges are fragile at $c = 1.96$, and at least one sign error occurs in every replication. This is not a symptom of poorly estimated individual judges; with many propensities around a common assignment-weighted average, some judge is likely to lie close to the boundary.

The simulations support the logic of the diagnostic. The population boundary distance d_z^0 is fixed within a design, sampling uncertainty enters through $\hat{d}_z$, and $\hat{r}_z$ measures how strongly the sample classifies the sign of d_z^0. Low values identify judges for whom sign mistakes are common; high values identify judges whose classification is stable. The calibrated

Table D.3: Design-level diagnostic performance

Design	$E[\hat{r}_{\min}]$	$E[\lvert\widehat{\mathcal{F}}(1.96)\rvert]$	Fragile share	Pr(any sign error)	$\Pr(\bar{p}^0 \in \widehat{\mathcal{I}})$
Stylized SM	0.796	0.950	0.317	0.372	1.000
Stylized AM	0.851	0.918	0.306	0.324	1.000
Philadelphia-like	0.259	5.774	0.679	0.778	1.000
Bail Philadelphia-like	0.640	2.060	0.229	0.513	1.000
Bail Miami-like	0.053	37.280	0.621	1.000	1.000

Notes: A judge is fragile when $\hat{r}_z \leq 1.96$. The fragile share is the average assignment share of fragile judges. The plausible boundary interval $\widehat{\mathcal{I}}(1.96)$ is the set of boundary locations that preserve the classification of every non-fragile judge. The final column reports the frequency with which it contains the true population boundary $\bar{p}^0$; this is not a coverage theorem, but it shows that the diagnostic-based interval does not exclude the true boundary in these designs.

designs show why this matters beyond the three-judge example: in many-judge settings fragility can arise even without targeting a single judge, which makes the diagnostic especially relevant in examiner and judge designs with many decision-makers. The diagnostic does not test average monotonicity and does not replace balance tests, first-stage checks, or exclusion discussions. It answers a different question: once average monotonicity is used to interpret the design, are the boundary classifications stably pinned down in the data?

E Additional Literature Discussion

This appendix positions the paper in the judge-IV and heterogeneous-IV literatures. The main text keeps the discussion short because the contribution is a boundary result, not a survey of examiner designs.

The first related literature studies the identifying assumptions behind judge and examiner designs. Frandsen et al. (2023) show that average monotonicity can preserve a proper weighted-average interpretation of the IV estimand even when strong monotonicity is too demanding. Chyn et al. (2025) and Goldsmith-Pinkham et al. (2025) provide methodological guidance for implementing examiner and leniency designs, and Sigstad (2026) measures monotonicity violations directly in judicial panels and studies when judge-IV estimates remain interpretable despite violations of strong monotonicity. This paper takes average monotonicity as the maintained fallback and asks whether it delivers a stable target population.

The second related literature studies IV estimands under treatment-effect heterogeneity. Imbens and Angrist (1994) show that monotonicity yields a local average treatment effect interpretation with a binary instrument. Mogstad and Torgovitsky (2024) emphasize that heterogeneous-IV interpretations can be sensitive to instrument structure and that average monotonicity is not purely behavioral. We use that observation to identify a specific discontinuity: under average monotonicity, the positive-weight response types switch when a judge propensity crosses the assignment-weighted average.

The paper also relates to work on testing judge-IV assumptions. Frandsen et al. (2023) propose a nonparametric test for the exclusion and monotonicity implications of judge designs, and Coulibaly et al. (2024) propose sharp testable implications and apply them to the Philadelphia bail setting. Those tests ask whether the identifying assumptions are rejected by the data. The boundary diagnostic asks a different question: conditional on using average monotonicity, does the estimated propensity profile stably classify the target population? The two are complements. Balance tests, first-stage reporting, exclusion arguments, and monotonicity tests address whether the design can identify a causal parameter; the boundary diagnostic addresses which population receives positive weight under average monotonicity and whether that population is stable around the estimated profile.

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